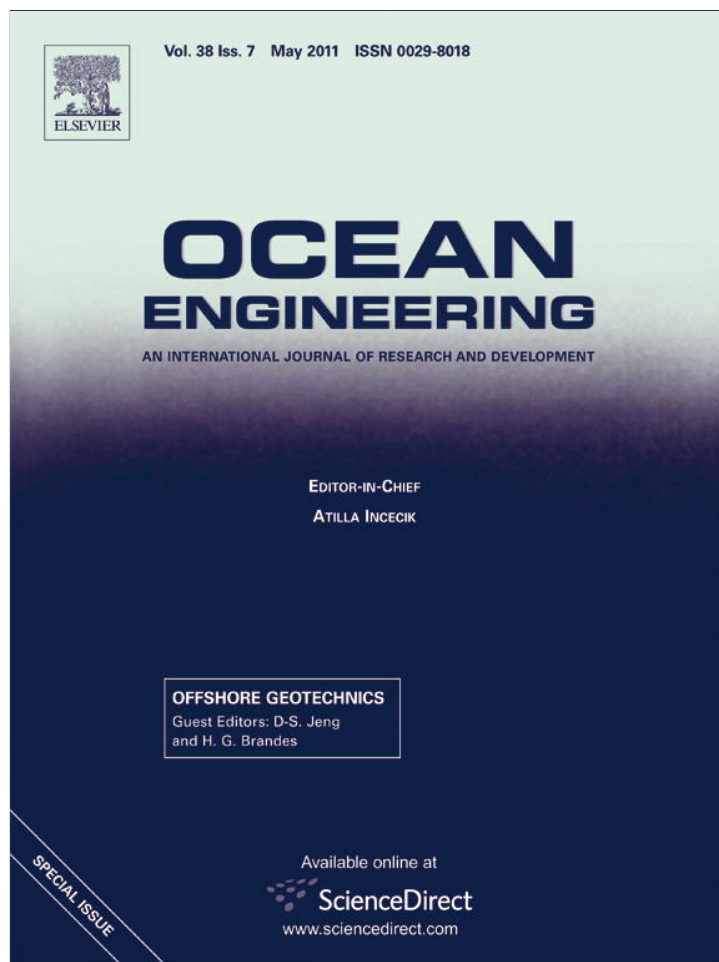




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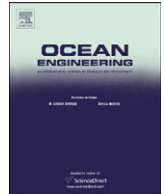
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Stability of submerged rock berms exposed to motion of liquefied soil in waves

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ABSTRACT

The paper describes the results of an experimental study on the behaviour of a submerged rock berm in liquefied backfill soil. The soil is liquefied by waves, and the rock berm is subject to the orbital motion of the liquefied soil. The soil used in the experiments was silt with $d_{50}=0.075$ mm. Various berm materials were used, stones of size 0.74–2.5 cm, plastic balls of size 3.6 cm, brass of size 2.5 cm and steel of size 1.0 cm. The experiments show that rock berms that are stable under very large waves can be unstable when they are exposed to the motion of liquefied soil. The limited data obtained in the study were plotted as a function of the mobility number versus the Keulegan–Carpenter number for the range of the Reynolds number of the tests. The critical mobility number corresponding to the incipient motion of the berm stones is determined. Recommendations are made as to how the present findings can be implemented in practice.

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1. Introduction

There are basically three kinds of protection measures against scour below pipelines (e.g., Sumer and Fredsøe, 2002): (1) the pipeline may be laid in a trench; (2) it may be covered with a stone protection layer (a rock berm); or (3) it may be covered with a protective mattress.

This paper is concerned with the first kind of protection. A viable option for this kind of stone protection is to place the pipeline in the trench, and subsequently install a rock berm over the pipeline (Fig. 1a). There are two scenarios related to this option. The trench and the rock berm are left open. However, in this case, the trench may be backfilled due to sediment transport (Fig. 1b), the natural backfilling; or, following the installation of the rock berm, the remainder of the trench is backfilled intentionally with the in situ sediment. The sediment in both cases (being in the loose state because the backfilling processes involve slow sedimentation) may be susceptible to liquefaction under waves. This may be due to momentary liquefaction (e.g., Sakai et al., 1992; Gratiot and Mory, 2000; Sumer and Fredsøe, 2002; Tørum, 2007; Michallet et al., 2009), or it may be due to residual liquefaction (e.g. Seed and Rahman, 1978; McDougal et al., 1989; Miyamoto et al., 2004; Sumer et al., 2006a; Damgaard et al., 2006; Sumer and Fredsøe, 2002; Jeng et al., 2007). This paper focuses on the residual liquefaction, caused by the buildup

of pore-water pressure. With the liquefaction of the sediment, internal waves emerge at the interface between the liquefied sediment and the water column (Miyamoto et al., 2004; Sumer et al., 2006a), and consequently the liquefied sediment experiences an orbital motion. Another likely scenario related to pipelines protected by a rock berm involves pipelines not placed in a trench but rather laid directly on the seabed. In this case the rock berm may be buried under migrating sand waves if the pipeline is crossing an active sand-wave area. Here, too, the sediment may be in the loose state, and therefore it may be susceptible to liquefaction under waves.

The conventional design strategy for a rock berm requires the stability of the top layer. For this, the Shields criterion is used; namely, the Shields parameter calculated for the top-layer stones must be smaller than the critical value of the Shields parameter corresponding to the initiation of motion. This is known as the Shields approach. Another approach (as an alternative to the latter) is to accept a certain amount of damage to the protection structure, an approach similar to that used in the design of rubble-mound breakwaters. This approach has been adopted by Klomp and Tonda (1995) and Tørum et al. (2008) to study the stability of rock berms. (It should be noted that allowing a certain amount of damage may avoid undue conservatism. However, the accepted damage should not lead to the failure of the structure.)

Although a substantial amount of knowledge on the stability of rock berms has accumulated over the years (see e.g., Sumer and Fredsøe, 2002, for a detailed account), as of yet, no study has investigated stability of such protection structures when they are exposed to the motion of *liquefied soil*, a situation described in the

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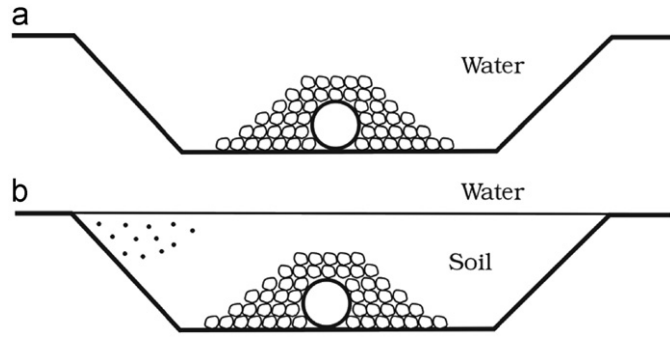


Fig. 1. (a) Installation of a rock berm over a pipeline. (b) The trench is backfilled (naturally or intentionally).

preceding paragraphs. The present study essentially addresses to this problem.

2. Experimental set-up

2.1. Wave flume and instrumentation

The experiments were carried out in a wave flume, 0.6 m in width, 0.8 m in depth and 26.5 m in length. Uniform, regular, progressive waves were produced by a piston-type wave generator. A wave absorber was used at the end of the flume to handle the reflection. The water depth was maintained at 40 cm.

The soil was placed in a 0.165 m deep, 0.56 m wide and 0.84 m long perspex box, located at 12 m from the wave generator. The box was placed in the flume so that the soil surface was flush with the false bottom of the flume (Fig. 2a).

In the tests, pore-water pressure and water-surface elevation (at the same section as the pressure measurements) were measured. The pressure measurements were made at the section $x=27$ cm from the offshore end of the box and at five depths, $z=0, 5, 10, 13$ and 16 cm, in which z is the vertical distance measured downwards from the mudline (Fig. 2a). The water-surface elevation was measured with a conventional wave gage. A few tests were carried out in the absence of the berm structure (and no pipe), and in these tests, the pressure and the water-surface elevation was measured at the centre of the silt box, $x=42$ cm.

The pore-pressure measurements were made, using Honeywell RS395 pressure transducers. The reader is referred to Sumer et al. (2010) for a detailed description of the instrumentation for the pressure measurements.

2.2. Soil

The soil used in the tests was silt. Fig. 3 displays two particle size distributions (PSDs) of the soil, one corresponding to the PSD before and the other after the present experimental campaign. As seen from Fig. 3, the finer fractions have been elutriated during the course of the experiments. The d_{50} , the grain size where 50% of soil is finer, is $d_{50}=0.075$ mm, the mean value found from the two PSDs, while the geometric standard deviation is $\sigma_g (= \sqrt{d_{84}/d_{16}}) = 1.65$, where the d_{84} and d_{16} are again the mean values based on the two PSDs.

The soil properties are given in Table 1. In the table, γ is the specific weight of water, γ_s the specific weight of sediment grains. The coefficient K_0 , the ratio between horizontal and vertical effective stresses, $K_0 = \sigma'_h / \sigma'_v$, was determined from $K_0 = 1 - \sin \varphi$ (Lambe and Whitman, 1969, p. 127) in which φ is the friction angle; the latter was obtained as the angle of repose in a simple test where the oven-dried soil was gently poured onto

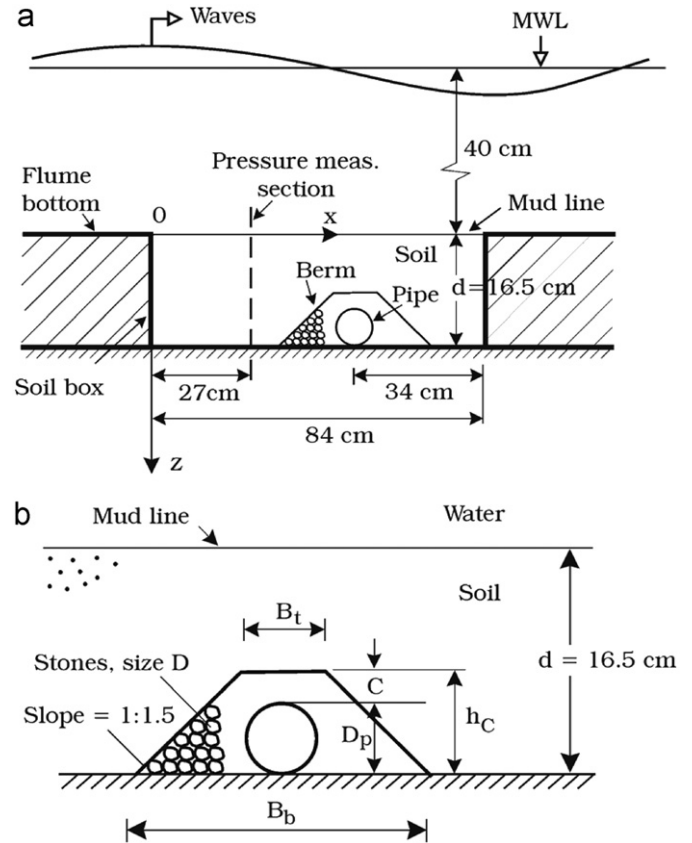


Fig. 2. Present experimental setup where the soil around the berm is liquefied by waves. (a) General; and (b) close-up of the rock berm (the cover). The figure is not to scale.

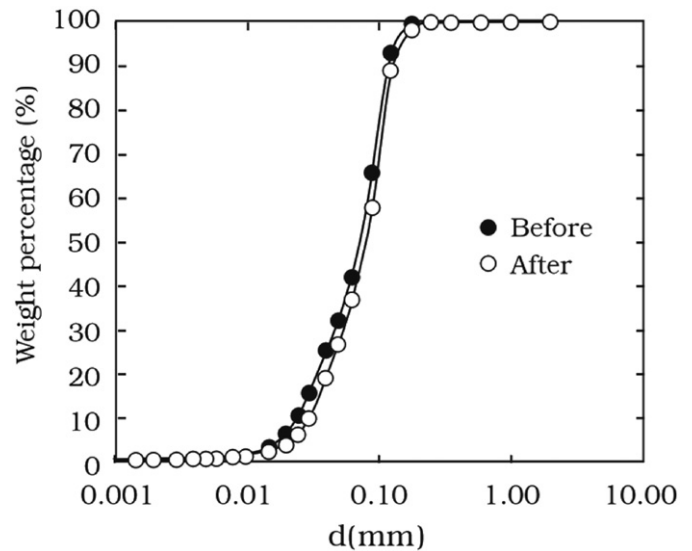


Fig. 3. Particle size distribution.

a horizontal plate in air, and the angle of repose was measured. The relative density

$$D_r = \frac{e_{\max} - e}{e_{\max} - e_{\min}} \quad (2.1)$$

is another soil property significant for liquefaction process (e.g., Sumer and Fredsøe, 2002). The values of the maximum and

Table 1
Soil properties.

Grain size, d_{50}	0.075 mm
Specific gravity of sediment grain, $s = \gamma_s/\gamma$	2.65
Ratio between horizontal and vertical effective stresses, K_0	0.50
Maximum void ratio, e_{max}	0.92
Minimum void ratio, e_{min}	0.47
Void ratio, e (before test)	0.75
Relative density, D_r (before test)	0.38
Total specific weight, γ_t	19.0 kN/m ³

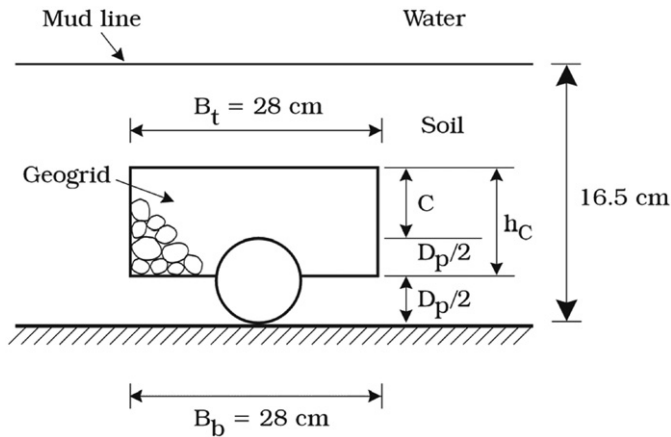


Fig. 4. Definition sketch. Geogrid case. The pipeline is protected by a geogrid.

minimum void ratios, e_{max} and e_{min} , and the value of the void ratio e were obtained in the same way as in Sumer et al. (2010). Table 1 also includes the total specific weight of soil, calculated from

$$\gamma_t = \frac{s+e}{1+e} \gamma \quad (2.2)$$

in which s is the specific gravity of soil grains.

2.3. Berm and pipe

In the majority of the tests, a trapezoidal-shaped berm was used (Fig. 2). Two supplementary tests (Tests 14–15) were carried out, however, with a cover with an approximately rectangular cross-sectional shape (Fig. 4) where the stones were enclosed in a geogrid, as will be detailed later.

The berm had a side slope of 1 in 1.5. Slopes milder than 1 in 1.5 were avoided not to interfere with the pore-pressure measurements (Fig. 2a). Also, the berm (with the pipe present) was placed not at the midway but slightly to the onshore side with an offset of 8 cm (Fig. 2a). This is also to avoid interference with the pore-pressure measurements.

Of the various cover material used, balls with $D=3.6$ cm (Table 2) were actually regular ping-pong balls. Their specific gravity, $s = \rho_s/\rho$, was adjusted so that it was close to the specific gravity of the liquefied soil, $s_{liq} = \rho_{liq}/\rho = 1.85–2.0$ (Sumer et al., 2006b), to observe the influence of the specific gravity of the cover material. This was achieved by placing metal pieces in the balls. The holes (through which metal pieces were inserted in the balls) were sealed when the desired specific gravity was obtained. As it was quite tedious to do this exercise, only a limited area of the top layer of the berm (namely an area of size 15 cm in the y -direction and 7.5 cm in the x -direction) was built with the balls and the rest was built from round stones with $D=2.5$ cm.

In the second case (Fig. 4), a geogrid was filled with $D=0.74$ cm stones. Two kinds of geogrids were tested. In the first case, the

geogrid was a net made from 0.2 mm wire, the mesh size being 1.2 mm (Test 14, Table 2). In the second case (Test 15, Table 2), it was an ordinary fabric, the mesh size being approximately 1 mm.

The model pipe was a 5 cm diameter and 53 cm long plastic pipe (the wall thickness being 3 mm). The two ends of the pipe were carefully sealed to ensure that there was no pressure dissipation. Although, in the majority of the tests, the pipe was held stationary in position (and therefore it was nothing more than a “dummy” cylinder), in two tests (Tests 8 and 9, Table 2), it was initially held free. In these latter tests, the pipe specific gravity was deliberately chosen to be $s_{pipe} = \rho_{pipe}/\rho = 1.2$, a small value, so that the pipe, despite the stabilizing cover weight, could float after the soil was liquefied, and therefore the effect of the upward motion of a floated pipe on the cover could be studied, as will be detailed later.

2.4. Velocity measurements

Two kinds of velocity measurements were made: (1) measurements of water velocity with no soil placed in the silt box (with and without the presence of the berm), and (2) measurements of liquefied-soil velocity (again, with and without the presence of the berm).

Regarding the measurement of water velocity in the absence of the soil and the berm (Test 17, Table 2), the water velocity was measured not only at the soil box section, but also outside the box, at a section 1 m onshore from the onshore end of the box, using a Laser Doppler Anemometer (LDA) equipment, a Dantec LDA-04 System with a 15 mW He–Ne laser (in forward-scatter mode). The velocity measurements were made under exactly the same test conditions as in the actual liquefaction experiments. Regarding the velocity measurements outside the soil box, clearly the rigid bed influences the particle velocity in the wave boundary layer (the thickness of the boundary layer being $O(1$ mm), Fredsøe and Deigaard, 1992). However, this influence will be nil for the particle velocities outside the boundary layer, and therefore the measured water-particle velocities with the rigid bed outside the boundary layer are identical to those experienced in the actual liquefaction experiments.

Velocity of liquefied soil was measured in two tests specially designed for this purpose where the motion of liquefied soil was videotaped from the side, in Test 3 in the absence of the berm, and in Test 6 with the berm present. In these tests, first the amplitude of the orbital motion of liquefied soil, a , at different soil depths was obtained from the video recordings, and then the amplitude of the orbital velocity was calculated from $U_m = 2\pi a/T$ in which T is the wave period.

2.5. Waves

The test conditions for the wave climate are summarized in Table 3. In the table, U_{m0} is the maximum value of the measured orbital velocity at the bed, U_{fm} is the maximum value of the friction velocity for the undisturbed case and calculated from

$$U_{fm} = \sqrt{\frac{f_w}{2}} U_{m0} \quad (2.3)$$

in which f_w is the wave friction coefficient and calculated from

$$f_w = \frac{2}{\sqrt{Re}} \quad (2.4)$$

in which Re , the wave-boundary-layer Reynolds number,

$$Re = \frac{a U_{m0}}{\nu} \quad (2.5)$$

where a is the amplitude of the orbital motion of water particles at the bed (Fredsøe and Deigaard, 1992). Furthermore, in Table 3,

Table 2
Test conditions.

Test no	Pipe	Cover material	Cover geometry					Remarks	
			Description	Size, D (cm)	Specific gravity	Cover thickness, C (cm)	Mean ^a top width, B_t (cm)		Mean ^a bottom width, B_b (cm)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
1	No pipe	No cover	-	-	-	-	-	-	Undisturbed case
2	No pipe	No cover	-	-	-	-	-	-	Undisturbed case
3	No pipe	No cover	-	-	-	-	-	-	Undisturbed case
									Liquefied-soil velocity distributions measured from video
4	Pipe fixed	Round stones	2.5	2.65	2.5	7.5	30	7.5	Compaction front was monitored
5	Pipe fixed	Round stones	2.5	2.65	5.0	7.5	37.5	10.0	Stone displacement measured
6	Pipe fixed	Round stones	2.5	2.65	5.0	7.5	37.5	10.0	Stone displacement measured
									Liquefied soil velocity from video
7	Pipe fixed	Round stones	2.5	2.65	10.0	7.5	52.5	15.0	Compaction front was monitored
8	Pipe free	Round stones	2.5	2.65	2.5	7.5	30.0	7.5	Stone displacement measured
9	Pipe free	Round stones	2.5	2.65	5.0	7.5	37.5	10.0	Stone and pipe displacement measured
10	Pipe fixed	Angular stones	0.74	2.65	5.0	7.5	37.5	10.0	Stone displacement measured
11	Pipe fixed	Angular stones	2.5	2.65	5.0	7.5	37.5	10.0	Stone displacement measured
12	Pipe fixed	Angular, crushed bricks	2.5	2.1	5.0	7.5	37.5	10.0	Stone displacement measured
13	Pipe fixed	Balls	3.6	2.1	6.0	7.5	40.5	11.0	Balls over the centre 15 cm width of cover, and the rest is 2.5 cm size round stones
									Ball displacement measured
14	Pipe fixed	Angular stones enclosed in geogrid made from 0.2 mm wire net, mesh size 1.2 mm	0.74	2.65	5.0	28.0	28.0	7.5	Geogrid/geotextile displacement measured
15	Pipe fixed	Angular stones enclosed in geotextile with mesh size 1 mm	0.74	2.65	5.0	28.0	28.0	7.5	Geogrid/geotextile displacement measured
16	Pipe fixed	Round stones in the absence of soil	2.5	2.65	5.0	7.5	37.5	10.0	Water velocity measured with LDA
		Cover is exposed to water							
17	No pipe	No stones, and no soil	-	-	-	-	-	-	Water velocity in the absence of soil and cover measured with LDA
									No displacement/damage
18	Pipe fixed	Brass prisms with hexagonal cross section. Height=2.5 cm, width=2.5 cm	2.5	4.0	5.0	7.5	37.5	9.5	
19	Pipe fixed	Steel circular cylinders. Height=1 cm, diameter=1 cm	1.0	7.4	5.0	7.5	37.5	9.5	No displacement/damage

See Fig. 2 for definition sketch for various quantities in the table.

^a Mean is taken over the length of the cover structure along the longshore axis.

Table 3
Test conditions for waves.

Wave height, H	17 cm
Wave period, T	1.6 s
Maximum value of orbital velocity at the bed; measured, U_{m0}	31.2 cm s^{-1}
Reynolds number, $\frac{aU_{m0}}{\nu}$	2.5×10^4
Maximum value of friction velocity, U_{fm}	2.5 cm s^{-1}
Grain Reynolds number, $\frac{dU_{fm}}{\nu}$	2.4
Shields parameter for the initiation of motion at the bed, θ_{cr}	0.08
Shields parameter, θ	0.5

Water depth is 0.40 m in all tests.

θ is the Shields parameter defined by

$$\theta = \frac{U_{fm}^2}{g(s-1)d_{50}} \quad (2.6)$$

in which g is the acceleration due to gravity.

2.6. Test procedure

The procedure regarding the placement of the soil in the flume was exactly the same as in our previous work (Sumer et al., 1999, 2006a). The procedure adopted in the tests including that of the placement of the berm stones is as follows. (1) Empty the flume at the end of the previous test. (2) Fill the soil box with water. (3) Place the pipe in the box and fixed it at the two ends so that it remains stationary during the test. (In the case of the free pipe (Tests 8 and 9), the pipe was held in place by a 2 mm diameter vertical rod, pressing on the pipe to avoid its floatation, the rod was gently removed immediately after the waves were switched on.) (4) Install the rock berm over the pipe. Measure the berm profile. (5) Place the soil into the water. (6) Fill the flume. (The duration from the beginning of Step 5 to the end of Step 6 was 90–120 min.) (7) Switch on the waves, and begin to sample continuously the water-surface elevation and the pore-water pressures. The sampling length was 15–20 min (which was large enough to cover the entire liquefaction–compaction process, Sumer et al., 2006a), and the sampling interval for the surface elevation and pressure recordings was 0.05 s. (8) Drain the water when the test ended. (9) With the water completely drained, excavate the soil to “unearth” the damaged berm, and measure the damaged berm profiles.

The measurement of berm profiles was made along the cross-shore and longshore axes indicated in Fig. 5. In a few tests, cross-shore profiles were also measured at other cross sections as well, parallel to the cross-shore axis indicated in Fig. 5. The way in which the cross-shore profiles were measured is as follows. Before the silt was placed in the soil box: top stones lying along the cross-shore axis were marked. Then, a string was stretched along the cross-shore direction at the top of the silt box between the offshore and onshore ends of the box, acting as a reference line. Subsequently, the (x,z) positions (see Fig. 2 for the coordinate axes) of the top of the marked stones were measured with a ruler with respect to the reference line. After the test was completed: the soil was excavated gently to unearth the berm. Subsequently, the same string was stretched in the same position as before, and then the (x,z) positions of the marked stones were measured. In this way, the (x,z) displacement of each and every marked stone was obtained (the damage). The longshore profiles were measured in the same way as in the cross-shore profiles.

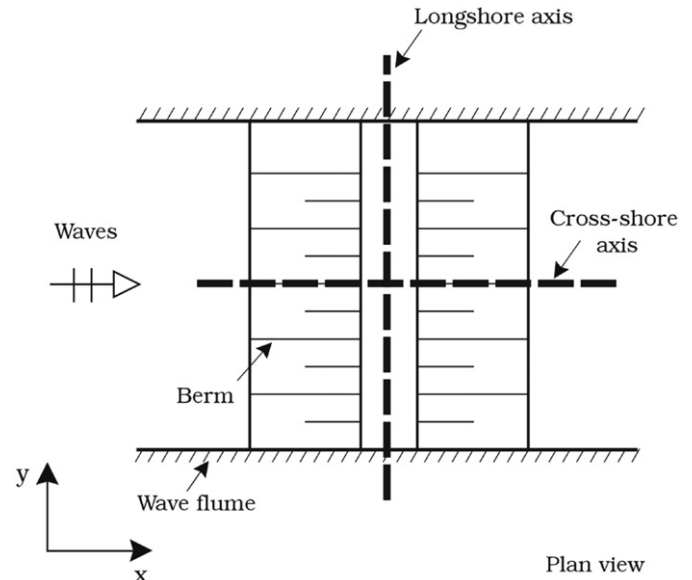


Fig. 5. Definition sketch. Plan view of the rock berm.

3. Sequence of soil behaviour during liquefaction

In the undisturbed case (with no berm and pipe present), three tests were carried out (Tests 1–3, Table 2), the reference case. These tests showed the following sequence of processes, in complete agreement with the previous research as described in e.g., Sumer et al. (2006a), Damgaard et al. (2006), and modelled by e.g., Seed and Rahman (1978), Sumer and Fredsøe (2002, p. 474) and Jeng et al. (2007): (1) With the introduction of waves, excess pore-water pressure builds up; (2) this is followed by liquefaction; (3) during liquefaction, the water column and the liquefied soil form a two-layered system of liquids (of different density), and this system experiences internal waves; (4) with the arrival of the liquefaction at the base bottom, compaction (and dissipation of the accumulated excess pore pressure) begins to occur, first starting at the impermeable base, and gradually spreading upwards; (5) upon the completion of the compaction process (by the time when the compaction front reaches the mudline), bed ripples begin to form on the surface of the soil.

In the presence of the berm structure, the previously mentioned sequence of liquefaction and compaction process occurs in much the same way. The pressure records showed that the behaviour of the pressure time series is exactly the same as in the undisturbed case (cf. Sumer et al., 2006a, Figs. 3 and 6). We do not give the time series in the present paper for reasons of space. Just to give an idea, the present data show that the soil at, for example, $z=16 \text{ cm}$ is liquefied after 11 waves. With the soil liquefied, the water column and the liquefied soil form a two-layered system of liquids, as described above, in the same way as in the undisturbed case. Our video recordings revealed that the interface experiences an internal wave, similar to the undisturbed case (Fig. 6).

Fig. 7 presents the data obtained from the video recordings (Test 6, Table 2) for the amplitude, a , of the orbital motion of liquefied soil at two sections, at $x=32 \text{ cm}$ (Section 1) from the offshore end of the soil box (square symbols), and at $x=50 \text{ cm}$ (Section 2), at the centre of the berm structure (crosses). Fig. 7 also includes the orbital amplitude data corresponding to the undisturbed case, with no berm, for comparison (filled circles). The a value for water particles obtained from the small-amplitude wave theory for a rigid bottom (potential theory) is also plotted in

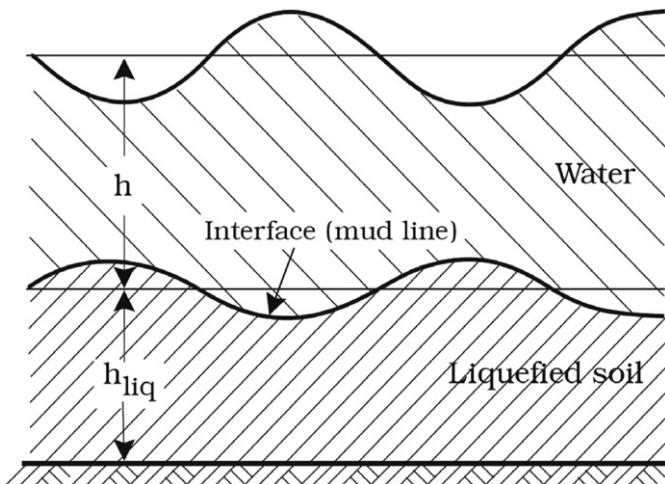


Fig. 6. Water column and the liquefied soil form a two-layered system of liquids (of different density), and this system experiences internal waves.

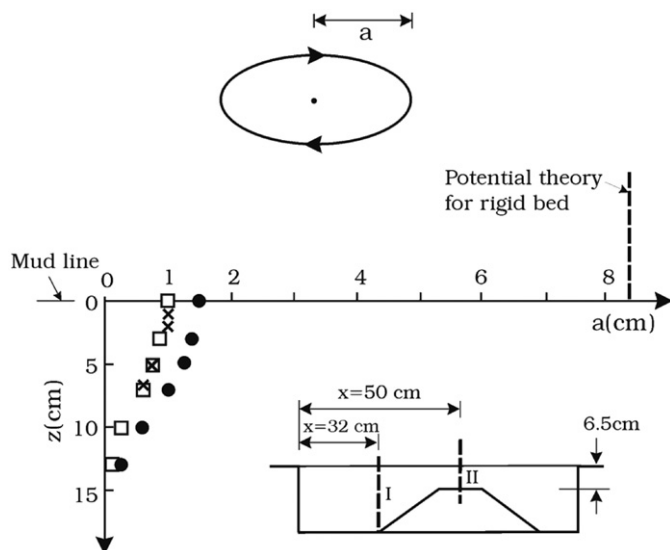


Fig. 7. Amplitude of liquefied soil as a function of depth. Square symbols: liquefied soil at $x=32$ cm; crosses: liquefied soil at $x=50$ cm (at the centre of the berm structure); and filled circles: liquefied soil, undisturbed case. Tests 3 and 6.

the figure as a reference line (dashed line). It should be noted that the amplitudes were measured for the stage where the soil was in the liquefied state across the entire soil depth. Fig. 7 shows that the amplitude a is practically the same for both Sections 1 and 2. However, it is clear from the figure that a is smaller than that experienced in the undisturbed case. This is essentially due to the fact that the presence of the berm structure “obstructs” the motion of the liquefied soil, and therefore the amplitude of the orbital motion of the liquefied soil is reduced.

The dissipation and compaction stage of the sequence of the soil behaviour is also qualitatively much the same as in the undisturbed case. Fig. 8 illustrates how the compaction front progresses with (Fig. 8b) and without (Fig. 8a) the presence of the berm structure. The origin of time, $t=0$, coincides with the instant where the first wave arrives at the pressure measurement section. The soil is liquefied first at the surface ($z=0$) within $t=0$ –6 s, and the liquefaction spreads down rapidly over a time period from $t=6$ to 18 s. Subsequently the compaction process begins where the soil grains come in contact with each other. The compaction process first begins at the impermeable base and gradually

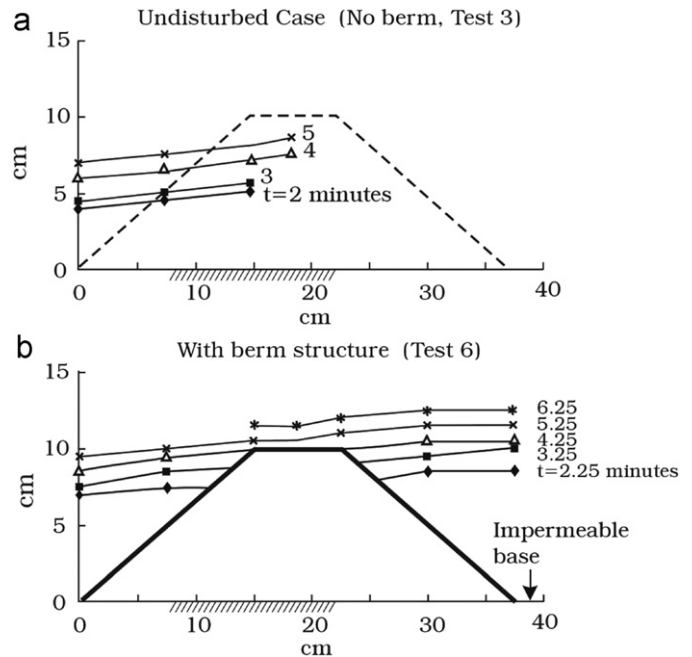


Fig. 8. Compaction front progresses with (a) in the absence of the berm (Test 3), and (b) in the presence of the berm (Test 6).

progresses in the upward direction, just as in the undisturbed case. (The compaction front from the video recordings could not be traced for times smaller than about 2 min because the viewing area of the video camera was blocked by the bottom horizontal supporting bar of the flume.) The slight tilt at the compaction front in Fig. 8a, the undisturbed case, is attributed to the fact that the liquefaction front reaches the impermeable base slightly earlier in the central areas, and therefore the compaction front will travel to slightly higher elevations in these areas. We also note that the physics of compaction process described in Sumer et al. (2006a) implies that the relative density of the compacted soil is expected to be similar to that in the undisturbed case, measured in our earlier study (2006a) as $D_r=0.78$ for similar wave and soil conditions.

Fig. 8 implies that the compaction spreads upwards more rapidly in the case of the presence of the berm. Although not shown here, the distribution of excess pore-water pressure with the depth was practically the same as that in the undisturbed case, meaning that the pressure gradient which drives the pore water upwards was practically the same as in the undisturbed case, and therefore the rapid progression of the compaction front in the case of the berm structure cannot be explained in terms of the driving pressure gradient. However, the presence of the berm structure in the soil “improves” the drainage, similar to the practice whereby gravel is introduced in the soil to help relieve the accumulated pore-water pressure. So the rapid progression of the compaction front in the case of the berm structure may be attributed to this effect. A consequence of this rapid progression of the compaction front may be that the cover stones will be exposed to the motion of the liquefied soil for shorter durations, and therefore leading to a lesser damage.

4. Stability of berm structure

4.1. General description

Fig. 9 presents the damage to the berm in Test 4, the solid line representing the cross-section of the berm structure before the

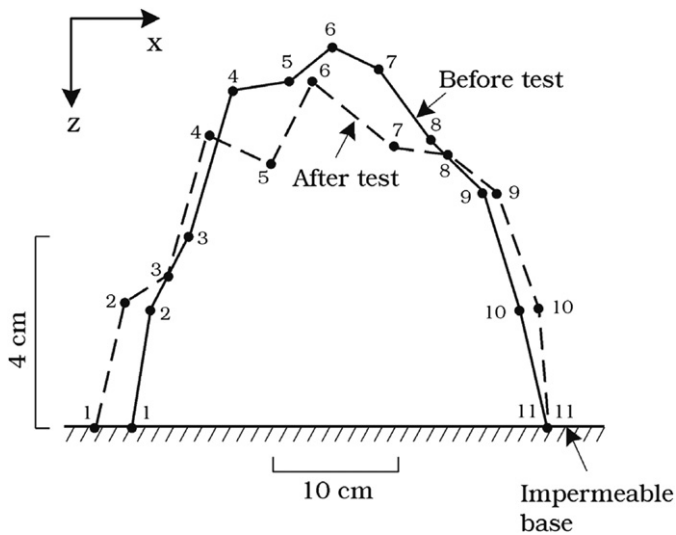


Fig. 9. Damage to the berm across the cross-shore direction. Test 4.

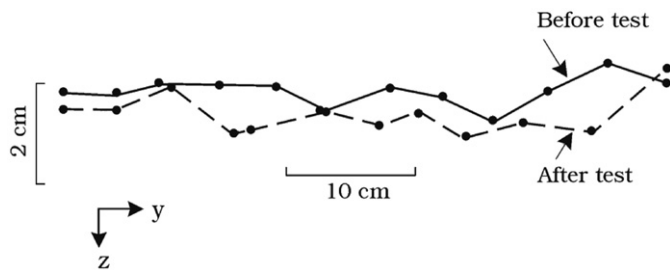


Fig. 10. Damage to the berm across the longshore direction. Test 4.

test, and the dashed line that after the berm structure is exposed to the orbital motion of the liquefied soil. Fig. 10, on the other hand, gives the damage across the spanwise length of the berm structure, i.e., along the longshore axis (Fig. 5) for the same test. The damage to the berm is fairly substantial. The crest of the berm experiences displacements of $O(1\text{ cm})$. The fact that the damage is uneven in the spanwise direction (Fig. 10) may be attributed to (1) the unevenness of the construction of the cover structure, and (2) the nonuniform shape and size of the stones. Although not shown here, the photographs of the berm structure before and after the test show that the stones during their displacement undergo not only translation but also rotation.

The displacement of stones (i.e., the damage) is due to the hydrodynamic forces on individual stones. There are two kinds of forces: (1) the agitating forces, the in-line force and the lift force, caused by the orbital motion of the liquefied soil, similar to those in the case when an ordinary berm structure is exposed to wave-induced flow in water; and (2) the stabilizing forces, the friction force and the submerged weight of stones. Obviously the stones were displaced because the resultant agitating force exceeded the resultant stabilizing force, as will be discussed in detail later.

Although the experimental constraints did not allow us to monitor the upward movement of the compaction front until it arrived at the level of $z \approx 7\text{ cm}$, Fig. 8b shows that the stones above this level were exposed to the liquefied soil for at least $O(2\text{ min})$ (the stones just at the top of the structure are exposed to the liquefied soil even longer, for $O(5\text{ min})$), and this indicates clearly that the stones were exposed to the orbital motion of the liquefied soil over a duration long enough, $O(2\text{--}5\text{ min})$, so that they can be displaced under the forces induced by the orbital motion of the liquefied soil.

4.2. Comparison with the case where the berm is exposed to orbital motion of water particles

Two special tests were carried out to compare the response of the berm structure when it is exposed to water with that when it is exposed to liquefied soil. In the first test (Test 16, Table 2), the berm structure was placed in position (Fig. 2a) without the soil, and therefore it was exposed to water in the test. In the second test (Test 5, Table 2), the soil was placed in the soil box and therefore the berm structure was, upon liquefaction, exposed to the liquefied soil. The berm structure was exactly the same in both experiments. The wave climate was also exactly the same in the two tests. The amplitude of the orbital velocity of water particles in Test 16, and that of liquefied soil particles in Test 5 (which was measured in Test 6, see Table 2) are plotted in Fig. 11 as a function of the vertical distance.

Although the orbital velocity in the water test is a factor of 7 larger than in the case of the liquefied soil (Fig. 11), stones did not move at all in the water experiment, and the berm structure remained completely intact (Test 16), while a fairly substantial amount of damage occurred (up to as much as $O(2\text{ cm})$ displacement in the vertical direction) in the case of the liquefied soil (Test 5).

First of all, to check for the incipient motion in the water case, we resort to the Shields diagram. To this end, the maximum value of the friction velocity is, from Eq. (2.3), $U_{fm,top} = \sqrt{f_w/2} U_{m,top} = 10.5\text{ cm/s}$. Here f_w , the wave friction coefficient, is calculated from $f_w = 0.308(a_{top}/k_s)^{-0.72}$ (Dixen et al., 2008; See also Kamphuis, 1975; Soulsby et al., 1993, for the wave friction coefficient) in which a_{top} is the amplitude of the orbital motion of water particles at the top of the cover structure, and taken as $a_{top} = U_{m,top}T/(2\pi)$, and k_s is Nikuradse's equivalent sand roughness for the stones and taken as $k_s = 2.5D$. The quantity $U_{m,top}$ is the amplitude of orbital velocity at the top of the cover structure and taken as 28 cm/s (Fig. 11). The above expression for f_w is valid for the range $0.5 < a_{top}/k_s < 10$ (Dixen et al., 2008), a condition satisfied by the present test conditions, a_{top}/k_s being 1.1. The calculated value of $U_{fm,top}$, namely $U_{fm,top} = 10.5\text{ cm/s}$, when inserted in Eq. (2.6) where d_{50} is replaced by D , the stone size, gives $\theta = 0.027$, smaller than $\theta_{cr} = 0.045$, the critical value of the Shields parameter for the initiation of the motion (see e.g. Sumer and Fredsøe, 2002, Fig. 1.2), and therefore the stones should not move in the water case, as revealed in the present water experiment.

The question is why the same stones under the same setting and under the same wave climate (with even a factor of 7 smaller orbital-velocity magnitude) move in the case of the liquefied soil.

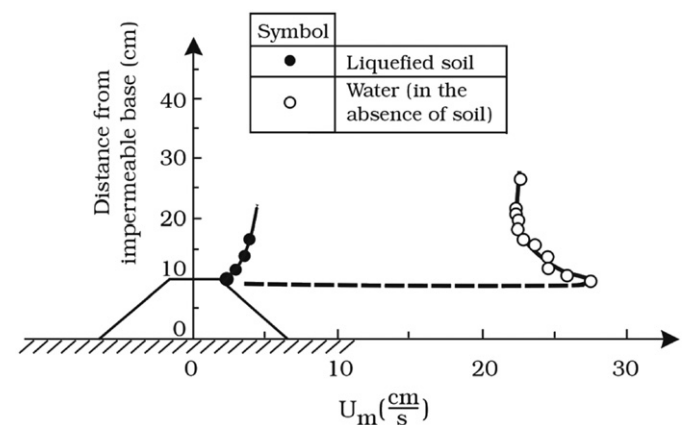


Fig. 11. Amplitude of velocity as a function of vertical distance. Water case, Test 16. Liquefied-soil case, Test 6.

The following factors contribute to the “earlier” incipient stone motion:

1. The submerged weight of the stones in water is $g(\rho_s - \rho)\pi D^3/6$ while it is $g(\rho_s - \rho_{liq})\pi D^3/6$ in liquefied soil, meaning that the stones are lighter in the liquefied soil by a factor of 2.3, considering $\rho_{liq}/\rho = 1.93$ (Sumer et al., 2006b) and $\rho_s/\rho = 2.65$ (Table 2).

2. This also implies that the weight and therefore the friction force (the force opposing to the motion of the stones) is a factor of 2 less in the case of the liquefied soil.

3. To the authors' knowledge, no study is available investigating forces on objects exposed to an oscillatory motion (or steady current, for that matter) of a liquefied soil. However, one can, to a first approximation, assume that stones experience an in-line force in the liquefied soil case, similar to the familiar Morison force (Sumer and Fredsøe, 1997). This force consists of two components, the drag component and the inertia component. The “composition” of these forces (the inertia and drag forces) is unknown as the liquefied soil behaves not as a Newtonian fluid. Making an analogy to the steady motion of a body in a fluid, Sumer et al. (1999) determined the resistance force and therefore the drag coefficient from an experiment where a “marine” object (a circular cylinder, or a sphere or a cube) falls in a liquefied soil. The latter data showed that the drag coefficient is 8–9 orders of magnitude larger than the water values, indicating that the drag force in the liquefied soil is expected to be larger in the liquefied soil.

4. Regarding the inertia component of the in-line force, this is composed of two parts: the apparent mass force and the Froude–Krylov force (Sumer and Fredsøe, 1997). The Froude–Krylov force should be increased by a factor of 1.93 in the case of the liquefied soil simply because the density of the liquefied soil is increased by a factor of 1.93. The second part, the apparent-mass force in the case of the liquefied soil, is unknown. Nevertheless this, too, is expected to increase because of the increase in the density of the liquefied soil.

5. A fifth contribution comes from the upward-directed pressure gradient $\partial p/\partial z$ in which p is the excess pore pressure (pore-water pressure in excess of the static pore-water pressure): this implies that the stones will be subjected to an additional lift force. (This lift force was discussed in conjunction with floatation of pipelines in liquefied soils in Sumer et al., 2006b).

From the above considerations, the agitating forces on the stones will be increased while the resistance force will be decreased in the case of the liquefied soil, resulting in an early threshold of the stone motion, as revealed by the present observations.

4.3. Degree of damage

Fig. 12 shows the damage in terms of the vertical displacement of the top stones, as a function of the cover thickness, C . The quantity S in Fig. 12 represents the average stone displacement in the vertical direction, calculated from

$$S = \left(\frac{\sum_{i=1}^N (\Delta z)^2}{N} \right)^{1/2} \quad (4.1)$$

in which Δz is the vertical component of stone displacement. The root-mean-square value is preferred in favour of the ordinary arithmetic mean to include in the average the negative values of the displacement, which occurred in a few cases. The average in the preceding equation is taken over the entire spanwise length of the berm structure where N was in the range $N=9-12$.

Fig. 12 shows that the damage, S , increases with increasing C , the cover thickness (see Fig. 2b for the definition sketch). This is because the top stones of the berm structure experiences larger and larger orbital velocities of the liquefied soil with increasing C (Fig. 7), and therefore the damage should also increase

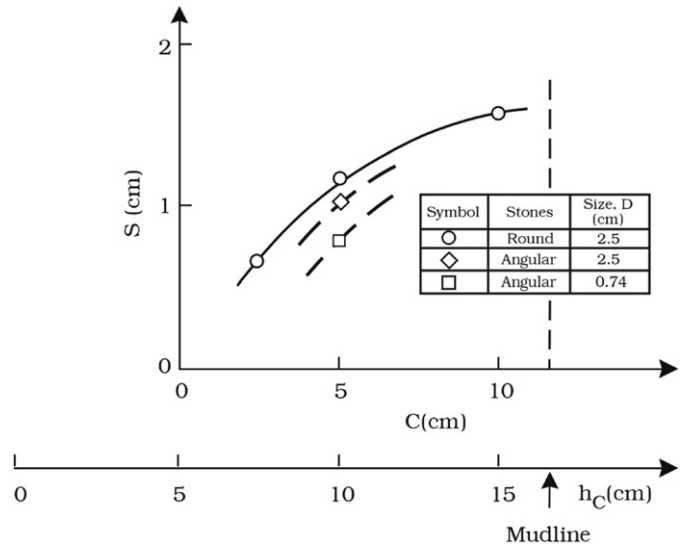


Fig. 12. Damage as a function of cover-layer thickness (see Fig. 2 for definition sketch).

with C . (At this juncture, we note the following knowledge for the case of breakwaters and regular rock berms. In breakwater tests and in tests related to stability of rock berms over pipelines, another definition of damage is given by

$$S_1 = \frac{A}{D_{n50}^2} \quad (4.2)$$

in which A is the area in cross section of the damaged part at the top of the breakwater or the rock berm, Tørum et al. (2008). The latter authors recommend that the damage S_1 should not exceed 50 for mounds covering pipelines, 2–3 m wide on top of the mound and with $D_{n50}=0.05-0.10$ m. For larger stones, the exceeding limit for S_1 should be lower.)

Now, returning to Fig. 12, the trend in the figure is such that $S \rightarrow 0$, as $C \rightarrow 0$. This is because when $C=0$, from Fig. 2, h_c becomes $h_c=5$ cm; this corresponds to a distance of $z=16.5-5.0=11.5$ cm, and the orbital amplitude (or alternatively the orbital velocity) corresponding to this value of z is, from Fig. 7, apparently very small. Hence, the damage will be practically nil.

The damage S seems to attain a constant value as C approaches the mudline ($C \rightarrow (d-D_p) = 16.5-5.0 = 11.5$ cm), the total depth of the soil. The figure shows that the maximum percentage damage is $S/h_c = (1.6 \text{ cm})/(15 \text{ cm}) = 0.1$.

For the same value of the cover layer thickness ($C=5$ cm), the damage S decreases when the stone shape is changed from round to angular. This can be explained by larger interlocking between the angular stones. In the angular stone case, the damage is smaller for smaller stones ($D=0.74$ cm, square symbol in Fig. 12) than for larger stones ($D=2.5$ cm, diamond symbol), and this can be explained by the sheer size of the stones; the smaller the stone size, the smaller the agitating forces, and therefore the smaller the displacement of the stone.

Fig. 13 shows the variation of the damage S with the specific gravity of stones, $s = \rho_s/\rho$ in which ρ is the density of water. Fig. 13 shows that the damage increases with decreasing s , as expected. This is particularly more pronounced for round stones, which is also expected.

4.4. Incipient stone motion in liquefied soil

As demonstrated in the preceding paragraphs, stones of a berm structure which cannot be moved in water can easily be moved in

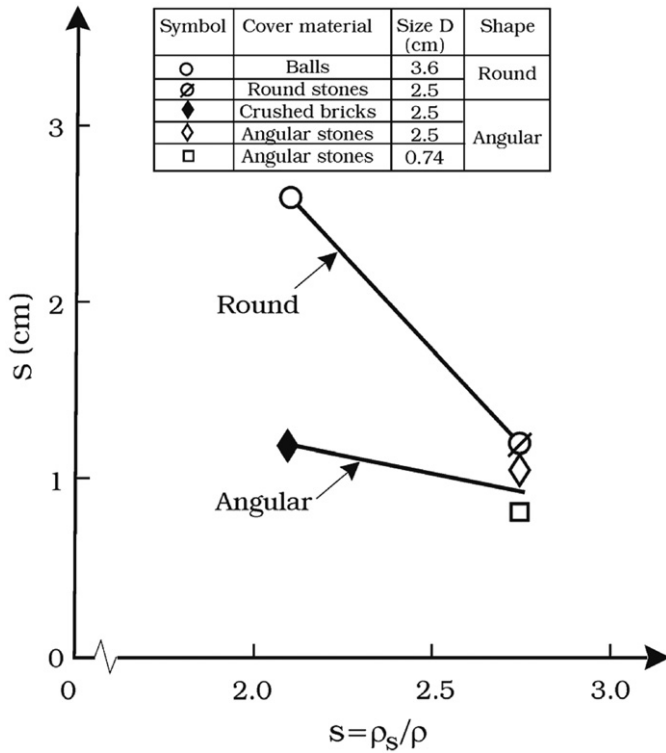


Fig. 13. Damage as a function of specific gravity of stones (see Fig. 2 for definition sketch).

liquefied soil under the same waves. The degree of the stone movement (i.e., the damage) in the latter case could be fairly substantial, and therefore the berm structure needs to be designed to ensure the stability of the structure if and when the soil is liquefied. The question is: What are the parameters which govern the initiation of stone motion in liquefied soil? This section attempts to address this question with the purpose to establish a design guidelines to determine the incipient motion.

On dimensional grounds, the initiation of stone motion is governed by three parameters, the so-called mobility parameter,

$$\Psi = \frac{U_m^2}{g \left(\frac{\rho_s}{\rho_{liq}} - 1 \right) D} \quad (4.3)$$

the stone Reynolds number,

$$Re_D = \frac{DU_m}{\nu_{liq}} \quad (4.4)$$

and the Keulegan–Carpenter number,

$$KC = \frac{U_m T}{D} \quad (4.5)$$

Here U_m is the amplitude of the orbital velocity of liquefied soil in the undisturbed case at the level of the top of the berm. The quantity ρ_{liq} is the density of the liquefied soil, and ν_{liq} the viscosity of the liquefied soil, which may be calculated from

$$\nu_{liq} = \frac{2}{2-3c} \nu \quad (4.6)$$

in which c is the solid concentration (volume concentration) of the liquefied soil (Cheng, 1997).

The first parameter can be interpreted as the ratio of the agitating force and the stabilizing force (i.e., the friction force, which is equal to $\mu(W-F_L)$ in which μ is the friction coefficient, W the submerged weight of the stone and F_L the lift force). We note that, in the above formulation, U_m is adopted in favour of the

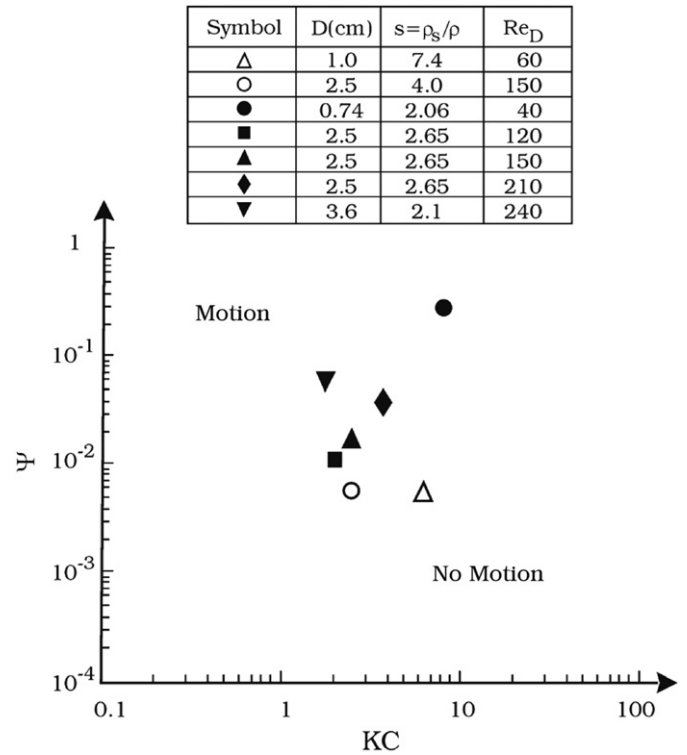


Fig. 14. Incipient stone motion in liquefied soil. Filled symbols: motion. Empty symbols: no motion.

friction velocity U_{fm} for simplicity. The above analysis can be considered as the extension of the classic Shields analysis to the incipient sediment motion in liquefied soil in waves, with the friction velocity replaced by U_m .

Fig. 14 shows the data plotted Ψ as a function of KC for the range of the Reynolds number experimented with, i.e., $40 \leq Re_D \leq 240$. The filled symbols represent the experiments where the berm stones moved whereas the empty symbols represent those where they did not move. The nondimensional quantities in the figure are based on U_m values which are calculated from $U_m = 2\pi a/T$ in which a values are taken from Fig. 7 for the undisturbed case. The specific gravity of the liquefied soil, ρ_{liq}/ρ , is taken as 1.93 (see Sumer et al., 2006b) while the concentration c in Eq. (4.6) is calculated from

$$\frac{\rho_{liq}}{\rho} = (1-c) + c \frac{\rho_s}{\rho} \quad (4.7)$$

The specific gravity of stones is given in the legend in Fig. 14.

The size of the data in Fig. 14 is too small to resolve the KC dependence of the critical mobility number for the range of the Keulegan–Carpenter number encountered in practice, namely $KC < O(40)$. However, the diagram can, to a first approximation, be used when KC remains within the range tested in the experiments, $1 \leq KC \leq 10$. It may be noted that the mobility-number data did not give any marked trend when plotted as a function of the Reynolds number. The real-life situations may involve large Reynolds numbers compared with the present Reynolds numbers. Therefore caution must be exercised when extending the results in Fig. 14 to prototype conditions. See the discussion under Section 4.7.

4.5. Influence of freely floating pipe

In two tests (Tests 8 and 9, Table 2), the pipe density was intentionally maintained at a very small value, $\rho_{pipe}/\rho = 1.2$ to

observe the influence of the pipe's floatation on the damage to the berm structure. The berm materials were round stones of size 2.5 cm. In Test 8, the cover thickness was $C=2.5$ cm while, in Test 9, it was $C=5$ cm. In both tests, the weight of the cover stones over the pipe was too small to prevent the pipe from floating. In Test 8, the pipe apparently broke through the berm structure, and was observed to float to the surface of the soil 1.5 min after the waves were switched on. In Test 9, the pipe floated only to a depth of 3 cm from the impermeable base. The pipe was found at this depth after the soil and top layers of the berm material were excavated upon the completion of the test. (No monitoring was made in this test as to the time development of the pipe's vertical position during its travel.) Stone displacements as much as $\Delta z = 3.5$ cm in Test 8 and $\Delta z = 2.5$ cm in Test 9 were observed (cf., the displacements plotted in Fig. 12 for the case where the pipe was held stationary). Clearly, with the pipe floating, the cover structure fails, and the result could be catastrophic, as demonstrated by these tests.

4.6. Geogrid test

Two tests were made with 0.74 cm stones enclosed in geogrids (Table 2, Tests 14 and 15). Geogrids offer an alternative solution to a rock berm. They improve the “drainage” characteristics of the soil in the same way as in the case of the berm structure. However, there are two advantages of geogrids: (1) they increase the dead weight attached to the pipeline; and (2) they maintain the integrity of the cover layer when exposed to liquefied soil under waves. These tests showed that (1) the excess pore-water pressure accumulated in exactly the same way as in the case of the stone cover; (2) the soil was liquefied, again, in the same manner as in the case of the rock berm; and (3) the geogrid remained “intact” during the sequence of liquefaction/compaction process, as expected.

Considering the damage experienced by the berm structure tested in the present experiments, this problem may be eliminated by confining the gravel in geosynthetic/geogrid layers (Mohri et al., 2001; Ling et al., 2003). The mesh size of the geogrid may be selected so that (1) the stability and integrity of gravel is ensured while (2) the drainage characteristics are not jeopardized.

4.7. Remarks on practical application

In order to apply the present results in a real-life problem, one needs to estimate the maximum value of the orbital velocity of liquefied soil in the undisturbed case at the level of the top of the berm structure, U_m . (This is to calculate the mobility parameter, Eq. (4.3) as well as the Keulegan–Carpenter number, Eq. (4.5)). To this end, the potential theory given in Sassa et al. (2001) who implemented Lamb's theory for wave propagation in a two-layer fluid for the present two-layered system may be used. However, comparison of the results from the latter theory and the present experimental results indicate that provisions need to be made in a typical field implementation to accommodate finite extent of the liquefied soil (in the offshore–onshore direction) to reduce the velocities predicted by the theory. With the above analysis, the procedure to implement the present results in a field situation may be summarized as follows. (1) Calculate the maximum value of the orbital velocity of liquefied soil, U_m , in the undisturbed case at the level corresponding to the top layer of the berm structure. (2) Calculate the mobility parameter (Eq. (4.3)). For the latter, use the information given in Sumer et al. (2006b) to assess the density of the liquefied soil, ρ_{liq} . (3) Calculate the Keulegan–Carpenter number from Eq. (4.5). (4) Calculate the Reynolds number from Eqs. (4.4) and (4.6). (5) Use Fig. 14 (to a first approximation), to check whether or not the designed stone size is large enough for the stone stability.

Clearly the prototype values of the Reynolds number will be large compared with those corresponding to the present tests. Unfortunately the way in which the Reynolds number influences the end result is unclear at the present time. Sarpkaya's (1975) water experiments for *free spheres* show that the force coefficients (the drag coefficient and the inertia coefficient) do not depend on the Reynolds number. Whether or not this result can be extended to the case of liquefied soil is unknown. Therefore caution should be observed when using the present results for field conditions. In this context, we also refer to the *steady-case* drag coefficient versus the Reynolds number variation for liquefied soils obtained for various geometrical shapes (circular cylinders, spheres and cubes) reported in Sumer et al. (1999). The latter study showed that the drag coefficient for very small Reynolds numbers ($Re_D \ll O(1)$) varies very significantly with the Reynolds number whereas, for large Reynolds number ($Re_D \gg O(1)$), no significant variation is observed. (Note that, here, the Reynolds number in Sumer et al., 1999, is converted to the present Reynolds number.)

If some damage is allowed to the berm structure, then Ψ can be allowed to be larger than the critical value for threshold (in Fig. 14). In this case, although it is at best suggestive, the model results given in Fig. 12 can be scaled up to make an assessment of the degree of damage. For this, (1) the damage S on the vertical axis in Fig. 12 should be scaled with h_c to get the percentage damage, and (2) the cover-layer height, h_c , on the horizontal axis should be scaled with d , the soil depth.

Finally we comment on the following three questions with regard to practical application of the present results.

The first question concerns the size of the sediment box in the experiments (Fig. 2). The sediment box of the present setup may be viewed as the model trench (although that was not the intention in the design of the test setup). Nevertheless, in this case the trench size in the model when scaled with the wave length will be relatively larger than in the prototype. Therefore the question is how the soil will behave in a narrower trench. The way in which a liquefied soil behaves in a trench as a function of the trench geometry is unknown, and this subject is worth exploring; a numerical study can be one way of doing this exercise, solving the differential equation governing the buildup of pore-water pressure coupled with the Biot equations. However, as far as the application of the key result (the incipient motion, Fig. 14) of the present study is concerned, the diagram in Fig. 14 can be used without any problem, since the quantities Ψ , KC , and Re_D in Fig. 14 are all based on the measured velocity corresponding to the undisturbed case at the level of the top of the berm. Provided that the end user determines this velocity in a real-life scenario, i.e., the approach velocity of the liquefied soil at the level of the top of the berm, then the application of the results will be straightforward.

The second question is that the soil used in the model tests is silt at “full” scale grain size. Therefore the question is how this soil behaves under full scale conditions. The idea of using the present silt is not to use the same size soil as in the prototype, but rather to use a soil which can be liquefied by the present wave loading. The similarity criterion for sediment is dictated not by the actual sediment size but rather by several other factors the most important of which are: (1) the sediment size should not be too coarse whereby all pore pressures developed in the sediment will dissipate as rapidly as they develop, resulting in no liquefaction; and (2) the sediment should not be too dense, so much so that there will be no room for the sediment grains to rearrange, again resulting in no liquefaction. These two criteria were the guidelines for us to select the particular sediment experimented with in the present study. We also note that another feature of sediment important for the application of the present results is the specific gravity of the liquefied sediment, ρ_{liq}/ρ . This is incorporated in

the mobility parameter, Ψ , and therefore accounted for in the nondimensional representation of the present results.

The third question is that the tests were carried out, using regular waves whereas waves are, for the most part, irregular in real life. Liquefaction process under irregular waves were studied earlier by the authors in conjunction with their work on sinking/floatation of marine objects in Sumer et al. (1999) (see also Sumer and Fredsøe, 2002). Although the data were not exhaustive, nevertheless the authors in the Sumer et al. (1999) study concluded that the combination of $H_s/\sqrt{2}$ and T_z gave the best agreement with the regular wave results in which H_s is the significant wave height and T_z the mean zero upcrossing period. In the same study the authors addressed questions such as the effect of history of wave exposure, the large time behaviour of the liquefied soil as well.

As a final remark, the authors believe that there is room for further research addressing the previously mentioned scaling issues. In this context, it is particularly important to conduct experiments similar in character to the present ones in a large-scale facility to test the present results against experiments with berms the size close to the field conditions as well as with much larger waves and periods close to field conditions.

5. Conclusions

A trench with a rock berm over a pipeline may be backfilled (naturally, or otherwise). If and when the backfill soil is liquefied, the rock berm will be subject to orbital motion of the liquefied soil. The experiments show that rock berms stable under large waves can be unstable when they are exposed to the motion of liquefied soil.

The parameters controlling the incipient stone motion under liquefied soil are the mobility parameter, Ψ , the Keulegan–Carpenter number, KC , and the Reynolds number, Re_D , (Eqs. (4.3)–(4.5), respectively). The limited data obtained in the present study were plotted as a function of Ψ and KC for the tested range of Re_D . These data indicate that the critical value of the mobility number corresponding to the incipient motion of stones is $\Psi_{cr} = (0.7–3) \times 10^{-2}$ for the range of KC and Re_D , $2 < KC < 10$, and $40 < Re_D < 240$, respectively.

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